

Development of a Robust Slippery Liquid-Infused Porous Surface with Grafted Polymer Brushes and Its Anti-Biofouling Applications in Marine Engineering

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Cite This: *ACS Appl. Polym. Mater.* 2023, 5, 5984–5994

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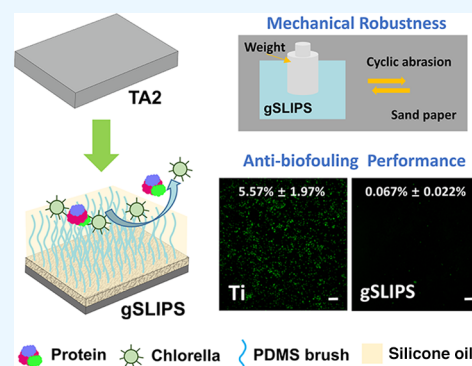
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ABSTRACT: It is difficult for any marine engineering equipment to avoid the impact of biofouling. Especially, for engineering equipment made of titanium with a high biocompatibility, biofouling is a thorny problem. The outstanding anti-biofouling properties of the slippery liquid-infused porous surfaces (SLIPSs) and the grafted polydimethylsiloxane (gPDMS) brush surfaces have attracted the attention of researchers. However, the durability and stability of the conventional SLIPSs and the wear resistance of gPDMS are insufficient and should be improved. In this work, PDMS molecule brushes were grafted onto a micro-arc oxidized porous surface infused with a lubricant (silicone oil) to develop a grafted SLIPS (gSLIPS) on a titanium alloy (TA2). The gPDMS molecule brushes have a stronger chemical affinity with silicone oil, which can preserve more lubricants to enhance the durability and stability of the SLIPS. Simultaneously, the infused lubricant in the micro-arc oxidized porous surface can improve the wear resistance of the grafted molecule brushes. Besides, the molecule brushes can further isolate the vulnerable titanium substrate from the biofouling microorganisms. By combining the advantages of the SLIPS and gPDMS, the gSLIPS has excellent stability, durability, and mechanical robustness. The gSLIPS possesses a better biofouling resistance than TA2 and gPDMS. For example, the coverage of *Chlorella* on the gSLIPS is $0.067\% \pm 0.022\%$ after being immersed for 14 days, which reduces by 98.8 and 95.6% compared with those on TA2 and gPDMS, respectively. In addition, the gSLIPS also has excellent anti-protein property. Therefore, this method will help develop a more stable, durable, and mechanically robust super-slippery coating with excellent anti-biofouling performance, which has great potential for the extensive titanium applications in marine engineering.

KEYWORDS: anti-biofouling, SLIPS, grafted polymer brush, polydimethylsiloxane, micro-arc oxidation



1. INTRODUCTION

Marine biofouling is the adhesion of undesired marine biofilms, which mainly consist of microorganisms, on the surfaces of materials.^{1,2} Marine biofouling can cause obstacles to marine, industrial, and medical applications. For example, the formation of biofouling on the surfaces of ships can increase the ship drag, corrosion, and fuel consumption. According to statistics, biofouling can globally cause an estimated loss of \$150 billion per year.³ Thus, many methods have been developed to mitigate the biofouling problem worldwide. Anti-biofouling coatings have played an important role in the biofouling control, for example, organotin-based coatings,⁴ zinc pyrithione coatings,⁵ and chlorothalonil coatings.⁶ However, many of the anti-biofouling coatings are toxic, and their damage to the marine creatures and environment cannot be ignored.⁷ Therefore, research on nontoxic and environment-friendly anti-biofouling coatings has become a hot spot in the marine engineering field.

In recent years, three types of environment-friendly coatings are considered to have great prospects for anti-biofouling purpose in marine engineering. The first kind of coating is the

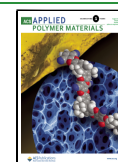
super-hydrophobic coating, which is formed by chemically modifying the surfaces with textures in micro/nano scale. Further, the water CAs on these surfaces are usually greater than 150° .⁸ This kind of coating can trap an air layer on the surface when being immersed into water to prevent the adhesion of the biofouling microorganisms.⁹ However, in the underwater environment, such coatings are prone to lose the air layer, and the surfaces are eventually adhered by biofouling microorganisms.¹⁰

The second type of coating is the SLIPSs (slippery liquid-infused porous surfaces) or super-slippery coating.¹¹ For example, Aizenberg et al. reported that a SLIPS could prevent up to 96–99% of common bacterial biofilm adhesion for at

Received: April 1, 2023

Accepted: July 11, 2023

Published: July 24, 2023



least 7 days in a low-flow rate environment.¹² Because of their excellent properties, the SLIPs are applied not only in anti-biofouling but also in many other fields, such as bubble transport,^{13,14} water collection,¹⁵ anti-icing,^{16,17} anti-corrosion,^{18,19} and so on. Meanwhile, many methods for preparing textured structures of the SLIPs have been developed, such as the laser etching process,^{20–22} electrochemical deposition,^{23–25} porous membranes,²⁶ layer-by-layer assembly,²⁷ and so forth. Although new application fields and preparation techniques of the SLIPs keep emerging, these methods are still complicated and uneconomical. Another main challenge for the SLIPs is to retain the lubricant and increase the stability and durability for a long-term service in the harsh marine environment.

The third type of coating is the grafted polydimethylsiloxane (gPDMS) brush surface fabricated by grafting liquid-like polymer molecules onto the solid surfaces via covalent bonds.²⁸ The solid surface is grafted with one end of the polymer molecule, while the other end can freely twist and wiggle. Thus, this type of coating also possesses dynamic surfaces, similar to the SLIPs.^{29,30} It has been proved that gPDMS can continuously prevent microorganisms and proteins from adhesion.³¹ If a biofouling microorganism cell or protein molecule touches the surface, the compressed polymer brushes will cause a spatial repulsion force and prevent its firm adhesion.³² However, this conventional gPDMS brush surface coating is more susceptible to wear damage, resulting in a relatively low robustness.³³ In addition, the majority of the current research is on grafting polymer brushes onto smooth surfaces. This is also a limitation for the grafted polymer brush coatings. Among the aforementioned three types of coatings, if the SLIPS and gPDMS could be combined together, they may overcome each other's shortcomings. This is because the polymer brushes could enhance the chemical affinity with the lubricant to preserve more lubricant for the SLIPS, and the existence of the lubricant could reduce the friction on gPDMS to increase its wear resistance and robustness.

Therefore, this work develops a SLIPS with grafted polymer brushes for a long-term, effective protection from biofouling. Linear methoxy-terminated PDMS molecules were grafted onto a micro-arc oxidized porous surface by a one-step equilibrium reaction. Subsequently, the sample was infiltrated with silicone oil to obtain a grafted SLIPS (gSLIPS). The gSLIPS has two characteristics to achieve its excellent stability, durability, and mechanical robustness. First, gPDMS has a better chemical affinity with silicone oil, which can lock the lubricant well through intermolecular forces.³⁴ Second, the lubricant infused into the porous surface can increase the wear resistance of the gSLIPS to improve its mechanical robustness. The high-rate centrifugation and long-term immersion test were conducted to enhance the capability to preserve the lubricant, and the improved mechanical robustness was proved by the abrasion test. To demonstrate the outstanding biofouling resistance of the gSLIPS, *Chlorella* and bovine serum albumin-fluorescein isothiocyanate (BSA-FITC) were used as the usual biofouling microorganism and protein. Therefore, this method will help develop a more stable, durable, and mechanically robust super-slippy coating with excellent anti-biofouling performance, which has great potential for the extensive titanium applications in marine engineering.

2. EXPERIMENTAL SECTION

2.1. Materials. Titanium alloy (TA2) raw samples were provided by GoldSky Titanium Metal Co., Ltd. (Hunan, China). Silicone oil (Aladdin, 20 mPa s) and sodium hydroxide were obtained from Aladdin Bio-Chem Technology Co., Ltd. (Shanghai, China). PDMS (Alfa, MW = 4000) was obtained from Beijing Innochem Science & Technology Co., Ltd. Glycerol (Macklin, Shanghai, China), sodium metaphosphate (Macklin, Shanghai, China), and sodium silicate (Damao, Tianjin, China) were obtained. Potassium fluoride was purchased from Guangzhou Chemical Reagent Factory. BSA-FITC (Solarbio) and phosphate-buffered saline (PBS, pH = 7.4) were obtained from Guangzhou Zhuanyan Biotechnology Co., Ltd. *Chlorella* was obtained from Xinrui Company (Zhuhai, China). Deionized water and artificial seawater were prepared in laboratory.

2.2. Preparation of gPDMS Anti-Biofouling Coatings. First, the porous surface (PS) was prepared with the same method as in our previous work.³⁵ Then, the prepared coatings were cleaned with toluene. Further, to obtain gPDMS, the PS was baked in vials containing PDMS for 24 h at 150 °C via a thermally activated reaction. The unreacted PDMS was removed from the sample with toluene. Finally, to obtain a gSLIPS, multiple silicone oil droplets (around 20 μ L) were introduced to the gPDMS. By maintaining the sample vertical for at least 1 h, additional silicone oil was removed.

2.3. Characterization. The surface morphologies of the coatings were characterized by FE-SEM (SU8010, Hitachi, Japan). The crystal structure of the coating was characterized by XRD (SmartLab, Rigaku, Japan) and Raman (inVia Qontor, Renishaw, UK) analyses. The chemical compositions of the samples were examined by an FT-IR spectrometer (Nicolet6700, Thermo Scientific, USA) and an XPS system (Nexsa, Thermo Scientific, USA). The contact angle (CA) and sliding angle (SA) were measured by a droplet shape analyzer (DSA100S, KRÜSS, Germany).

Our previous work's approaches were used to characterize the stability, durability, and anti-biofouling property.³⁵ The anti-protein properties of the samples were characterized by a protein adhesion test. BSA-FITC was selected as the experimental protein. The samples were immersed in 25 μ g mL⁻¹ BSA-FITC in PBS solution for 8 h in the dark at 25 °C. The samples were retrieved after 8 h and washed three times with PBS solution to eliminate the poorly adsorbed BSA-FITC. CLSM was used to detect the adsorbed BSA-FITC on the sample surfaces.

3. RESULTS AND DISCUSSION

3.1. Wettability of gSLIPS Coatings. As shown in Figure 1, a SLIPS with grafted polymer brushes for a long-term, effective protection from biofouling was developed. In short,

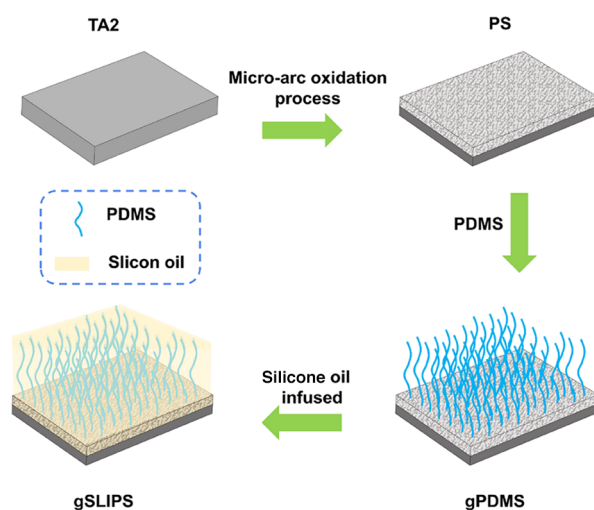


Figure 1. Schematic of the preparation of the gSLIPS coating.

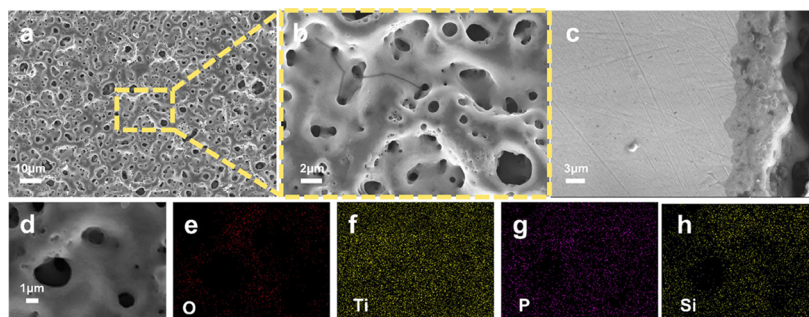


Figure 2. Surface morphologies of anti-biofouling coatings. (a) SEM image of the PS morphology. (b) Enlarged SEM image of the region shown in (a). (c) Cross-sectional SEM image of the PS. (d) Selected area for EDS mapping. (e–h) EDS mappings for O, Ti, P, and Si elements of the corresponding area, respectively.

linear trimethylsiloxy-terminated PDMS molecules were grafted onto the PS to get the gPDMS. Subsequently, the sample was infilled with silicone oil to obtain the gSLIPS. The wettabilities of the samples were measured by water CA (Figure S1). It can be seen that the surface wettability of the samples changes from hydrophilic to hydrophobic. It is noteworthy that the CA of the gSLIPS is smaller than that of the gPDMS. In theory, the chemical composition and the surface roughness of the sample determine its wettability.^{36,37} gPDMS is formed by grafting a thin PDMS molecular layer onto the rough micro-arc oxidized surface. Therefore, the roughness of gPDMS is similar to the rough micro-arc oxidized surface. However, the gSLIPS is obtained by locking a relatively thick silicone oil layer, which can cover the rough micro-arc oxidized surface to form a very smooth surface. As the chemical composition of both the gPDMS and gSLIPS is mainly siloxane, the difference in wettability between the gPDMS and gSLIPS could be determined by the surface roughness. Based on the experimental results, the CA of the gPDMS with a rough surface is larger than that of the gSLIPS with a smooth surface, which is consistent with the Wenzel equation.³⁸

3.2. Surface Characterization. It can be seen from the SEM image that the PS is made up of many pores with diameters in micro-/nanometer scale (Figure 2a,b). Further, there are also many pores inside or between the large holes, which is a common porous foam construction.³⁹ The PS coating has a structure consistent with the results of other researchers, consisting of a porous layer and an inner dense layer (Figure 2c).^{40,41} As shown in the EDS patterns in Figure 2e,f, the main elements in the PS are Ti, Si, P, and O. TiO₂ rutile and TiO₂ anatase phases are included in the phase composition of the PS, which were characterized by XRD and Raman spectroscopy analyses (Figure S2).^{40,42–44}

Figure 3a shows the chemical reaction diagram of grafting PDMS onto the PS. The gPDMS was obtained by a one-step reaction of trimethylsiloxy-terminated PDMS with the PS without any solvent, catalyst, and other conditions. In short, trimethylsiloxy-terminated PDMS reacted with the hydroxyl groups on the PS through two possible mechanisms.^{45,46} Then, the chemical composition of the PS, gPDMS, and gSLIPS are characterized by FT-IR analysis (Figure 3b). The peak at 2962 cm^{−1}, which corresponds to the peak of −CH₃, can be seen in the gPDMS spectrum. This characteristic organic alkane (−CH₃) peak is from PDMS, indicating that PDMS was successfully grafted onto the micro-arc oxidized PS.^{47,48} In addition, it can be noted from the FT-IR spectrum that the

new peaks appear in the spectrum of the gSLIPS. The peaks at 1260 and 1410 cm^{−1} are the symmetric and antisymmetric vibrational peaks of −CH₃ in Si−(CH₃)₂, respectively. Further, the peaks at 2905 and 2962 cm^{−1} are the characteristic peaks of −CH₃.^{49,50} These results demonstrate that the gPDMS is covered by silicone oil.

Furthermore, the XPS results exhibit that the Ti 2p peak disappeared in the spectrum of the gSLIPS compared to that of the PS and gPDMS (Figure 3c). The main XPS peaks in the gSLIPS are O 1s, C 1s, and Si 2p, which indicate its main elements. The O 1s and Si 2p XPS spectra of the PS, gPDMS, and gSLIPS are depicted in Figure 3d–i. The peaks of O 1s at 535 eV (± 0.5 eV) and 532 eV (± 0.5 eV) are attributed to O–Si.⁵¹ Further, the peak (530.5 eV (± 1 eV)) corresponds to −OH in the PS and gPDMS, which could be the hydroxyl group on their surfaces.⁵² The Si 2p XPS spectra are shown in Figure 3g–i, with the peaks at 105.4 eV (± 0.5 eV) and 102.5 eV (± 0.5 eV) attributed to Si–O.^{53,54} The peaks located at 103.04 and 102.75 eV correspond to C–Si in the gPDMS and gSLIPS, respectively.⁵⁴ In addition, the atomic ratio and ratio of Si/O on the PS, gPDMS, and gSLIPS are shown in Table S1. For the PS, the O and Si elements come from SiO₂ and TiO₂ (or SiO₂) in the micro-arc oxidation layer, respectively. Hence, the atomic ratio of O/Si in the PS should be more than 2, which is consistent with the results. Normally, the Si and O elements on the gPDMS and gSLIPS are mainly from the PDMS molecule (or silicone oil molecule). The theoretical atomic ratio of O/Si in the PDMS molecule (or silicone oil molecule) is close to 1. Thus, the atomic ratio of O/Si on the gSLIPS or the gPDMS is close to 1, which indicates that PDMS was formed on the PS.

3.3. Surface Free Energy of Anti-Biofouling Coatings.

In general, a good wettability of a droplet on the surface indicates a high surface free energy, while a poor one results in a low surface free energy.⁵⁵ CA or surface tension can be used to calculate the surface free energy.⁵⁶ For example, the surface free energy of the prepared sample surface can be obtained by testing the CAs of the water and diiodomethane droplets (WCA and DCA, respectively) on the solid surface with the following equations:^{57–59}

$$\gamma_{LV}(1 + \cos\theta) = 2(\gamma_S^d \gamma_L^d)^{1/2} + 2(\gamma_S^p \gamma_L^p)^{1/2} \quad (1)$$

$$\gamma_S = \gamma_S^d + \gamma_S^p \quad (2)$$

Here, γ_{LV} is the interfacial free energy; θ is the CA of the liquid on the sample surface; γ_S and γ_L are the solid surface free energy and the liquid surface free energy, respectively; γ_S^d is

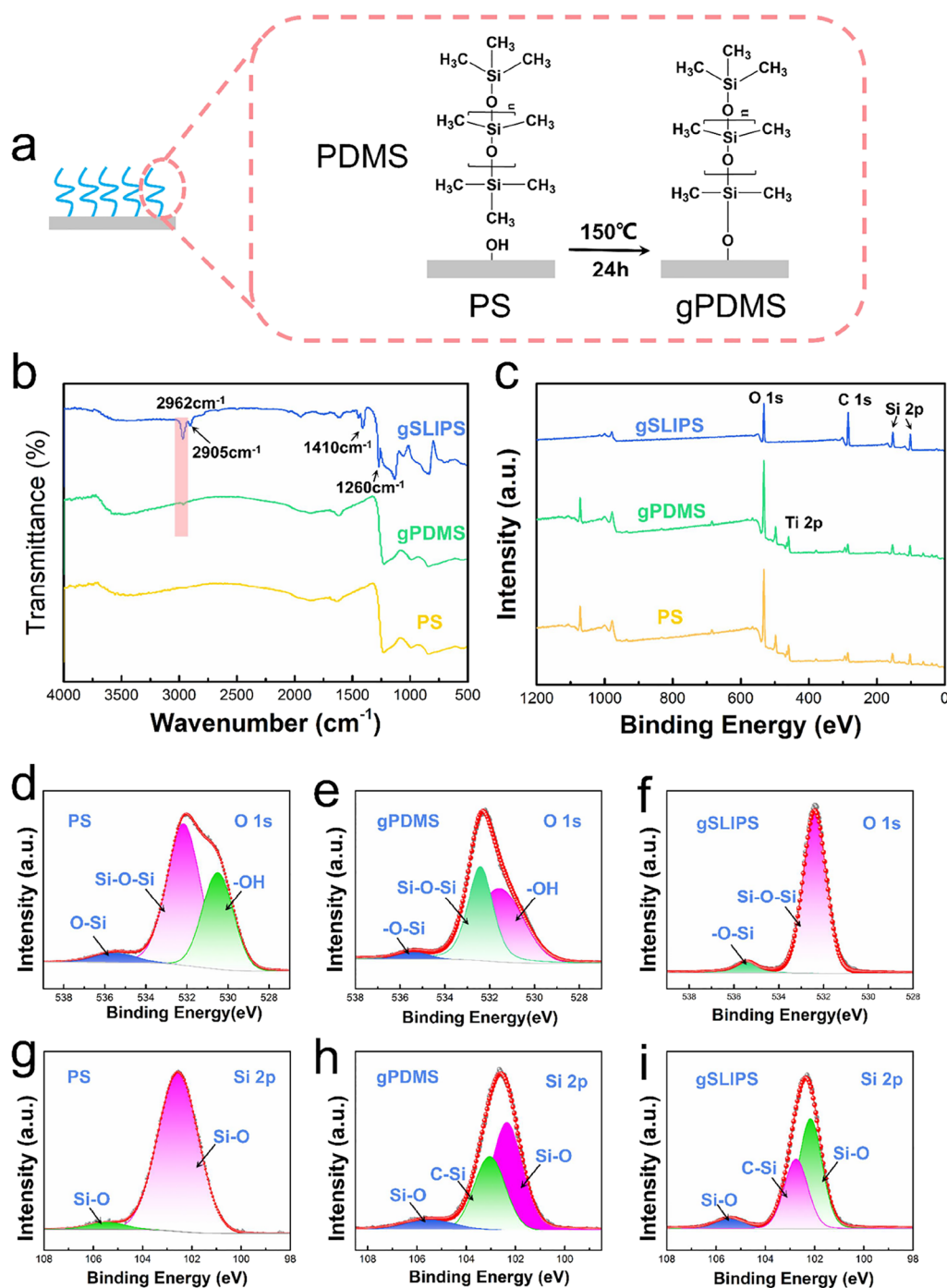


Figure 3. Chemical composition characterizations of anti-biofouling coatings. (a) Schematic illustration and chemical reaction diagram of grafting the polymer brushes onto the PS. (b) FT-IR spectra of the PS, gPDMS, and gSLIPS, respectively. (c) XPS spectra of the PS, gPDMS, and gSLIPS, respectively. (d–i) O 1s and Si 2p XPS spectra of the PS, gPDMS, and gSLIPS, respectively.

the solid surface free-energy dispersion component; and γ_s^p is the solid surface free-energy polar component. The dispersion component of the liquid surface free energy is denoted by γ_L^d , and γ_L^p is the liquid surface free-energy polar component. To calculate γ_s of different samples, it is necessary to measure the CAs of water and diiodomethane on different sample surfaces, respectively. The CAs of water and diiodomethane on the sample surfaces are shown in Figure S3a. The calculated surface free energies for the sample surfaces are shown in Figure S3b and Table 1.

Table 1. Calculated Surface Free Energies for Different Samples

samples	TA2	gPDMS	gSLIPS
WCA ^a (deg.)	70.4 ± 1.8	110.3 ± 2.2	107.5 ± 1.5
DCA ^b (deg.)	29.9 ± 2.3	89.9 ± 3.3	80.8 ± 0.4
γ_s^d (mN m ⁻¹)	38.564	11.628	16.160
γ_s^p (mN m ⁻¹)	7.673	1.277	0.897
γ_s (mN m ⁻¹)	46.237	12.905	17.057

^aWCA: water contact angle. ^bDCA: diiodomethane contact angle.

It can be observed that the WCAs and DCAs of the gPDMS and gSLIPS are higher than that of TA2. The WCAs of TA2, gPDMS, and gSLIPS are $70.4^\circ \pm 1.8^\circ$, $110.3^\circ \pm 2.2^\circ$, and $107.5^\circ \pm 1.5^\circ$, respectively, and the DCAs of TA2, gPDMS, and gSLIPS are $29.9^\circ \pm 2.3^\circ$, $89.9^\circ \pm 3.3^\circ$, and $80.8^\circ \pm 0.4^\circ$, respectively. In addition, TA2 has the highest surface free energy of 46.237 mN m^{-1} , while the surface free energies of the gPDMS and gSLIPS are 12.905 and 17.057 mN m^{-1} , respectively. The gSLIPS has a little greater surface free energy than the gPDMS, which has the lowest surface free energy.

Various studies have shown a nonmonotonic relationship between the adhesion force of the biofouling film and the surface free energy of the anti-biofouling coatings. As the critical surface tension for the spreading of the interfacial water layer on the liquid substrate is 22 mN m^{-1} ,⁶⁰ the coatings with a surface free energy of $20\text{--}25 \text{ mN m}^{-1}$ show the best capability for anti-biofouling.^{61,62} The surface free energy of the gSLIPS is near the aforementioned range. Thus, the gSLIPS has better anti-biofouling properties, compared with TA2 and gPDMS.

3.4. Stability, Durability, and Mobility of Anti-Biofouling Coatings. Besides the surface free energy, the mobility of the liquid on the super-slippy coating surfaces is another key indicator for evaluating the anti-biofouling properties of the super-slippy coating. As shown in Figure 4a,b, the mobilities of the samples are assessed by testing the

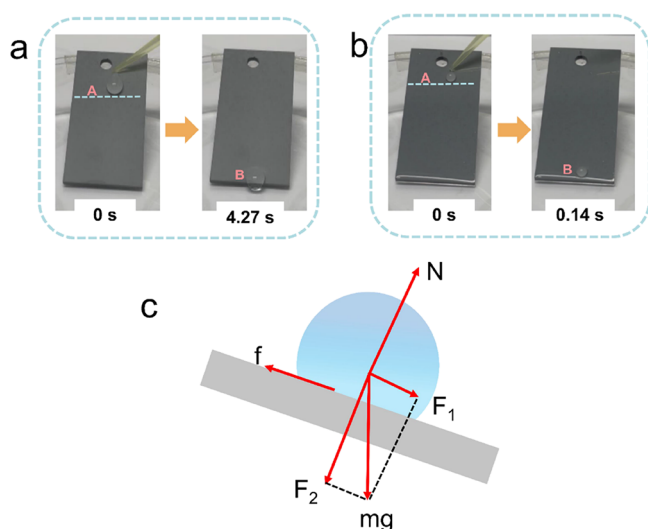


Figure 4. Sliding processes of the water droplets on different coatings: the (a) gPDMS and (b) gSLIPS with the droplet volumes of 75 and $2 \mu\text{L}$, respectively. (c) Force analysis of the droplet sliding process.

sliding time of a water droplet on the sample surfaces tilted at an angle of 18° . To differentiate the sliding movement of the water droplets on the surfaces, the sliding experiment was performed by adding 75 and $2 \mu\text{L}$ water droplets on both the surfaces of the gPDMS and gSLIPS (Figure S4 and Table S2). Then, each sliding process was recorded. The $75 \mu\text{L}$ water droplet slides from A to B on the gPDMS within 4.27 s, while the $2 \mu\text{L}$ water droplet slides the same distance on the gSLIPS in only 0.14 s. The results show that the water droplet slides faster on the gSLIPS than it does on the gPDMS. Figure 4c depicts the force analysis of the droplet sliding process. The mobility of the droplet is influenced by two forces: the static friction between the surface and the droplet (f) and the

component of gravity along the surface (F_1).⁶³ Droplets start to slide on the surface only when $F_1 > f$, and the droplet sliding speed on the surface is associated with the properties of the material surface and the droplet volume. Usually, the droplet with a larger volume will slide faster on the substrate surface.⁶⁴ It can be seen that whether the droplet volume is 2 or $75 \mu\text{L}$ on the surfaces, the sliding time of the gSLIPS is remarkably less than that of the gPDMS (Figure S4 and Table S2). It indicates that the gSLIPS has a much lower sliding resistance. Generally, the lower the sliding resistance of the coating surface, the easier it is to remove biofouling from the coating surface.

An accelerated lubricant loss test (high-rate centrifugation test) was conducted to examine the coating's capability to retain the lubricant to explore the stability of the gSLIPS. The gSLIPS was put in a high-speed centrifuge and centrifuged at different speeds (0–12,000 rpm) for 1 min. The stability of the sample was characterized by testing the SAs and CAs of the sample before and after centrifugation. As shown in Figure 5a, both the SAs and CAs on the sample surfaces changed after centrifugation. The CAs on the sample surfaces after centrifugation increase as the centrifugation speed increases. Centrifugation leads to the loss of silicone oil, which exposes the surface of the gPDMS, and the CA on the gPDMS surface is slightly larger than that on the gSLIPS. Therefore, it causes an increase in the CA on the sample surface after centrifugation. However, the SA of the gSLIPS remains below 10° even at 12,000 rpm. In contrast, the SA is $23.9^\circ \pm 3.9^\circ$ for the SLIPS at 12,000 rpm (Figure S5). It indicates that the grafted polymer brushes on the gSLIPS can lock the lubricant well due to its better chemical affinity with the silicone oil, resulting in its excellent stability.

In addition, durability is also an important feature for anti-biofouling coatings. To investigate the durability of the gSLIPS, the CA on the gSLIPS was characterized by the immersion experiment with artificial seawater. As shown in Figure 5b, after the immersion in artificial seawater for 0 day and 7 days, the water CAs on the gSLIPS are $107.5^\circ \pm 1.5^\circ$ and $107.8^\circ \pm 0.6^\circ$, respectively. In addition, the change of the CA on the gSLIPS is less than 0.5° after the immersion in artificial seawater for a week. Even with the gSLIPS immersion in artificial seawater for 3 weeks, the CA is $107.9^\circ \pm 0.5^\circ$; the change of the CA on the gSLIPS is still less than 0.5° . Furthermore, the chemical tolerance of the gSLIPS was characterized by measuring the CA after the immersion in strong acid solutions for several days (Figure S6a). After the immersion in the strong acid solution ($\text{pH} = 1$) for 3 days, the CA only decreases by 1.7° compared to that before the immersion. Even after 5 days of immersion in the strong acid solution, the CA decreases by 4.7° . In addition, Figure S6b depicts that CAs of the gSLIPS are higher than 105° for the droplets with pH values from 1 to 14. The above results show that the gSLIPS has an excellent durability.

3.5. Mechanical Robustness of Anti-Biofouling Coatings. Mechanical robustness is also important for anti-biofouling coatings in real applications. Therefore, the abrasion test was used to assess the wear resistance of the gPDMS and gSLIPS coatings (Figure 6). The samples were placed on 1500 mesh sandpapers, loaded with 400 Pa pressure, and abraded reciprocally on the sandpaper (Figure 6a,b). After various abrasion cycles, the samples were positioned at an 18° angle (Figure 6c) to observe and record the movement of the water droplet. For easy observation, $75 \mu\text{L}$ and $2 \mu\text{L}$ water drops

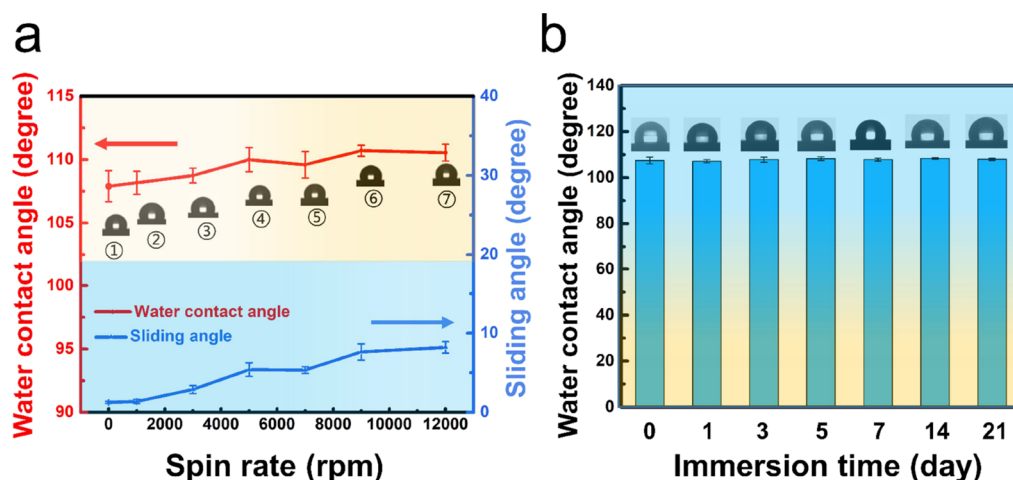


Figure 5. Stability and durability of the gSLIPS. (a) SAs and CAs of the gSLIPS after a high-rate centrifugation. (b) CAs of the gSLIPS after the immersion for various days.

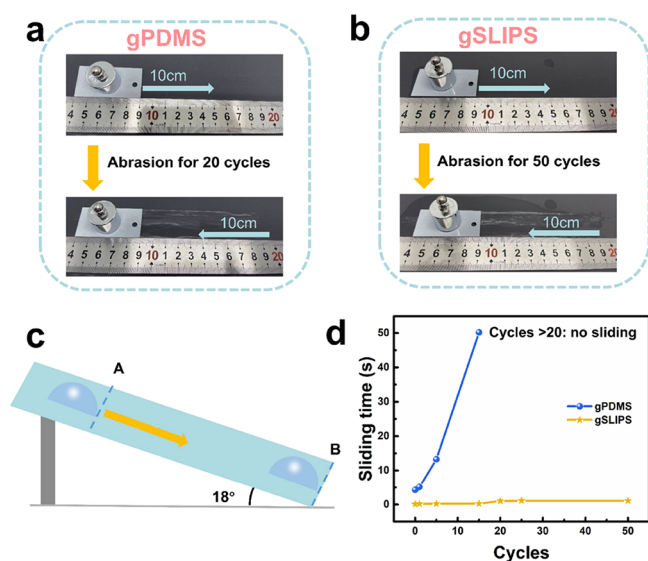


Figure 6. Mechanical robustness of the coatings. Photos of the abrasion tests for the (a) gPDMS and (b) gSLIPS loaded with 400 Pa pressure on a 1500 mesh sandpaper, respectively. (c) Schematic diagram of a water droplet sliding on the surface of the sample tilted at an angle of 18°. (d) Sliding times of the droplets from Point A to Point B on the surfaces of the samples after different abrasion cycle numbers.

were added onto the surfaces of the gPDMS and gSLIPS, respectively. After 20 cycles of abrasion, the gPDMS left large gray debris on the sandpaper surface. The droplet is anchored on the gPDMS and cannot slide to Point B (Figure S7b1). After 50 cycles of abrasion, the droplet on the gPDMS shows a pendant shape. The droplet is anchored at the starting point and cannot slide (Figure S7b2). The results investigate that the wear resistance of gPDMS is poor.

On the contrary, after 50 cycles of abrasion, the droplets on the gSLIPS can still slide rapidly from Point A to Point B within only 1.07 s (Figure S7c). Unlike the scenario of the gPDMS, the sliding times of the droplet on the gSLIPS are all short after different abrasion cycles, not exceeding 1.5 s (Figure 6d). The above results reveal that the PDMS molecules on the gSLIPS exhibit liquid-like characteristics, and the droplet can slide on its surface. After being infused with the lubricant, the

wear resistance of the gSLIPS can effectively improve. The results show that the gSLIPS has excellent mechanical robustness.

3.6. Anti-Biofouling Performance of Anti-Biofouling Coatings. *Chlorella* was selected as the typical biofouling microorganism to evaluate the biofouling resistance of coatings. TA2, gPDMS, and gSLIPS were individually immersed with *Chlorella* for the anti-biofouling testing. CLSM was used to characterize the attachment of *Chlorella* on the sample surface (Figure 7a–c), and the corresponding coverage values are shown in Figure 7d. It is not difficult to find that after immersion for 1 day, many *Chlorella* are already attached on the surface of TA2. Similarly, there are a little uneven *Chlorella* attaching on the surface of the gPDMS, while it is difficult to observe *Chlorella* attaching on the gSLIPS. In Figure 7d1, after immersion for 1 day, *Chlorella* covered $1.63\% \pm 0.79\%$ of the area on TA2. Compared with that of TA2, the coverage of *Chlorella* on the gPDMS and gSLIPS for 1 day immersion decreases by 69.3% and 99.4%, respectively.

Further, the coverage of *Chlorella* on the TA2 surface arise $4.37\% \pm 0.45\%$ and $5.57\% \pm 1.97\%$ after 7 and 14 days of immersion, respectively. For the gPDMS, the coverage *Chlorella* is $0.5\% \pm 0.69\%$, $1.11\% \pm 0.75\%$, and $1.53\% \pm 0.73\%$ for 1, 7, and 14 days, respectively (Figure 7d). The above result indicates that the coverage of *Chlorella* on the gPDMS reduces by 72.5% compared to that of TA2 for 14 days' immersion. This is because the polymer brush on the gPDMS surface has a liquid-like property and can isolate the biofouling.

Notably, for 1 day immersion, $0.0092\% \pm 0.0041\%$ area of the gSLIPS was covered by *Chlorella*. Even after immersion for 14 days, the coverage is barely $0.067\% \pm 0.022\%$. The coverage of *Chlorella* on the gSLIPS for 14 days reduces by 98.8 and 95.6% compared to those of TA2 and gPDMS, respectively. The anti-biofouling results indicate that the gSLIPS has an excellent performance against algae adhesion compared to TA2 and gPDMS.

At the initial stage of biofouling formation, it is mainly composed of biomacromolecules, such as proteins, called condition film.⁶⁵ The condition film is a necessary prerequisite for the formation of biofouling. Therefore, it is important to explore the coating's resistance to protein adhesion. The antiprotein capabilities of the samples were evaluated by

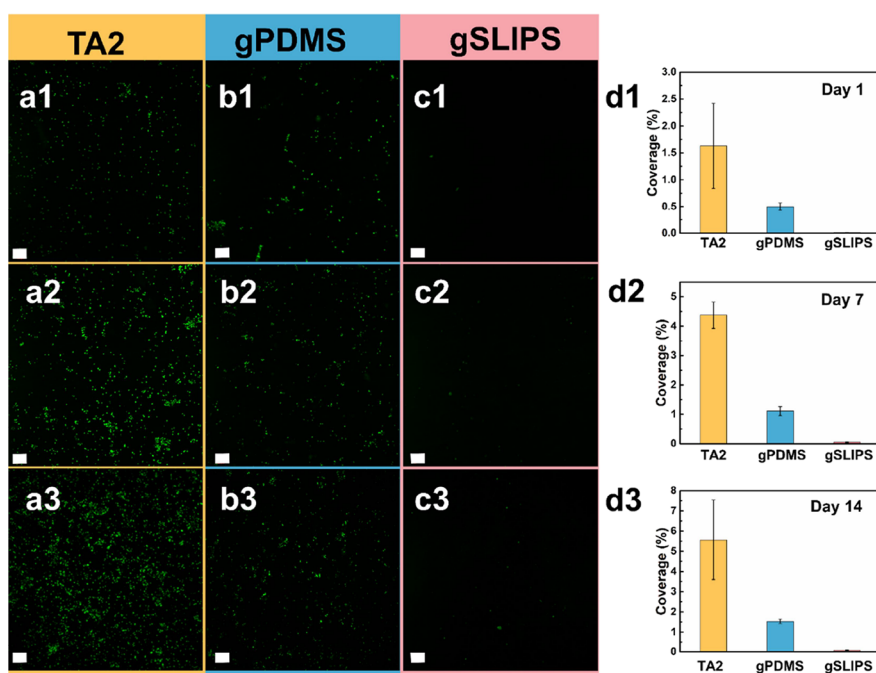


Figure 7. Anti-biofouling tests of the coatings with biofouling microorganisms. CLSM images of (a1–a3) TA2, (b1–b3) gPDMS, and (c1–c3) gSLIPS after immersion with *Chlorella* for 1, 7, and 14 days, respectively. (d1–d3) The corresponding coverage values of *Chlorella* on TA2, gPDMS, and gSLIPS after immersion for 1, 7, and 14 days, respectively (Scale: 100 μm).

analyzing the coverage of BSA-FITC on the sample surfaces. First, TA2, gPDMS, and gSLIPS were immersed in a BSA-FITC solution, respectively. After the immersion in the dark at 25 $^{\circ}\text{C}$ for 8 h, the samples were removed and dried under natural exposure in the dark. Then, the FITC-BSA coverage on the sample surfaces was observed by CLSM.

The TA2 surface exhibits a significant amount of green fluorescence, indicating that BSA-FITC is extensively adsorbed on the TA2 surface, as shown in Figure 8a. The coverage statistics show that the coverage percentage of BSA-FITC on the TA2 surface reaches $35.59\% \pm 9.73\%$, as shown in Figure 8d. The coverage percentage of $22.66\% \pm 0.31\%$ of BSA-FITC on the gPDMS is less than that of TA2 (Figure 8d). This is because the polymer molecules on the gPDMS can form a physical barrier for protein adhesion. In addition, only a little fluorescence is observed on the gSLIPS surface, with a coverage percentage of $1.29\% \pm 0.46\%$. The coverage percentage of BSA-FITC on the gSLIPS reduces by 96.4 and 94.3% compared to those of TA2 and gPDMS, respectively. It indicates that the gSLIPS could effectively reduce the adhesion of BSA-FITC.

3.7. Anti-Biofouling Mechanism of Anti-Biofouling Coatings. Figure 9 displays the possible anti-biofouling mechanism based on the results of the biofouling-resistant tests for TA2, gPDMS, and gSLIPS. There are significant biofouling issues because of the high biocompatibility of titanium alloys to attract microorganisms. As the polymer brush on the gPDMS has a liquid-like property and can isolate biofouling, the coverage percentage of biofouling on the gPDMS is less than that on TA2. However, the coverage of *Chlorella* on the gPDMS increases linearly with the increase of the immersion time. After the immersion in the *Chlorella* culture media for 14 days, the coverage of *Chlorella* on the gPDMS is threefold more than that on the gPDMS for 1 day. This could be because the density of the PDMS molecules on the gPDMS is lower, which allows the biofouling (e.g.,

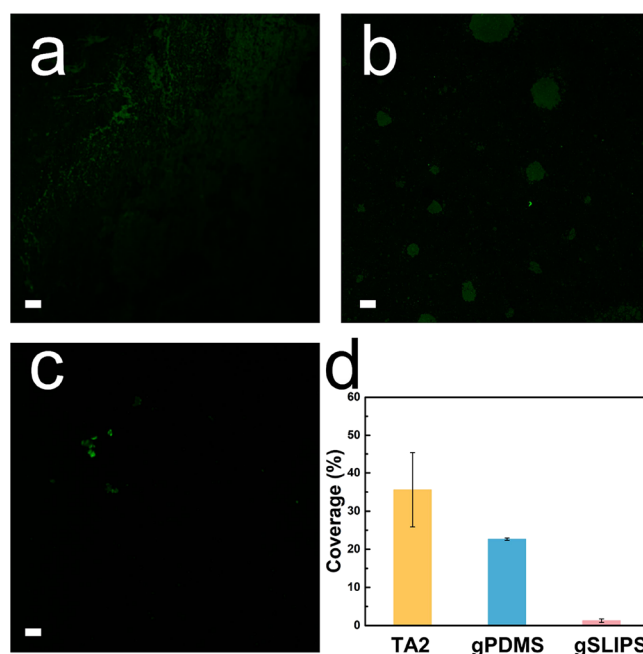


Figure 8. Anti-biofouling tests of the coatings with proteins. CLSM images after the FITC-BSA adhesion tests for different coatings: (a) TA2, (b) gPDMS, and (c) gSLIPS. (d) Statistical results of the FITC-BSA coverage on different coatings (Scale: 40 μm).

Chlorella or protein) to penetrate through the polymer brush and adsorb onto the substrate (Figure 9a).³²

Apparently, the long-lasting biofouling resistance of the gSLIPS is attributed to the lubricant and liquid-like PDMS molecule brushes. First, the PDMS molecules on a micro-arc oxidized porous surface has a better chemical affinity with the lubricant, which can lock the lubricant well through the intermolecular forces. These distinctive characteristics can

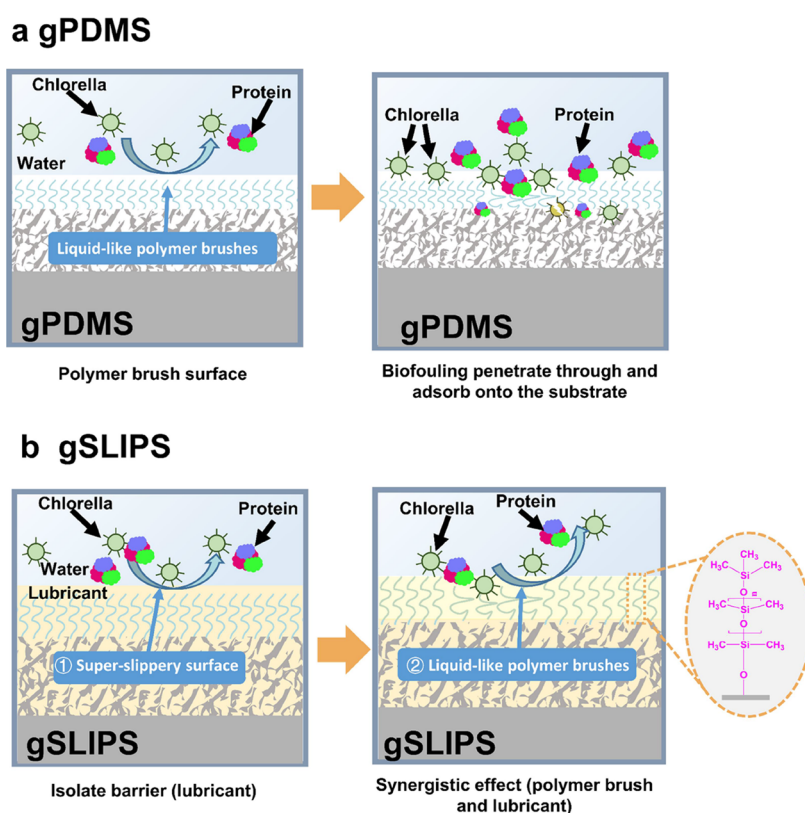


Figure 9. Schematic illustration of the anti-biofouling mechanism of the (a) gPDMS and (b) gSLIPS.

guarantee the prolonged storage and preservation of the lubricant, which ensure the outstanding stability and durability of the gSLIPS. Second, the lubricant layer is changeable and mobile. This super-slippy surface can hinder the initial attachment of the microorganisms. The microorganisms appear to keep floating when in contact with the lubricant. For a long-term immersion, even if biofouling attempts to pass through the lubricant layer due to the gravitational sedimentation of microorganisms or their appendages, the PDMS molecule brushes in the lubricant layer will prevent their attachment to the substrate surface. Because the microorganisms or their appendages can compress the polymer brushes, it would cause spatial repulsion and prevent biofouling from firmly attaching. Eventually, by combining the advantages of the above mechanisms, the significant biofouling resistance of the gSLIPS on the titanium alloy can be achieved.

4. CONCLUSIONS

In summary, we have developed a SLIPS with grafted polymer brushes (gSLIPS) for a long-term, effective protection from biofouling. The gSLIPS combines the advantages of the SLIPS and the grafted molecule brushes. The gSLIPS has two characteristics to achieve its excellent stability, durability, and mechanical robustness. First, the gPDMS has a better chemical affinity with silicone oil, which can lock the lubricant well through intermolecular forces. Second, the lubricant infused into the porous surface can enhance the mechanical robustness of the molecule brushes. Besides, the molecule brushes can further isolate the vulnerable titanium substrate from the biofouling microorganisms.

The coverage of *Chlorella* on the gSLIPS is $0.067\% \pm 0.022\%$ after being immersed for 14 days, which reduces by 98.8 and 95.6% compared to those of TA2 and gPDMS,

respectively. In addition, the gSLIPS also has excellent resistance to the protein adhesion. The coverage percentage of BSA-FITC on the gSLIPS reduces by 96.4 and 94.3% compared to those of TA2 and gPDMS, respectively. This outstanding biofouling resistance can be attributed to several anti-biofouling mechanisms, resulting from the combination of the super-slippy surface and grafted polymer brushes. This work reveals that the grafted polymer brushes improve the stability, durability, and anti-biofouling capabilities of the gSLIPS. Therefore, this method will help develop a more stable, durable, and mechanically robust super-slippy coating with excellent anti-biofouling performance, which has great potential for extensive titanium applications in marine engineering.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsapm.3c00667>.

CAs of TA2, gPDMS, and gSLIPS; XRD patterns and Raman shift spectra of the PS; surface elemental concentrations of the PS, gPDMS and gSLIPS; surface free energy and wettability of anti-biofouling coatings; sliding processes of the water droplets on different coatings; sliding time comparison between the gPDMS and gSLIPS with different volumes of water droplets on their surface; SAs of the gSLIPS and SLIPS after the high-rate centrifugation; chemical tolerance of the gSLIPS; and photos of water mobility on the surfaces of the gPDMS and gSLIPS with different cycles of abrasion (PDF)

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Author Contributions

H.Y. and X.J. conceived the research. D.Y., X.J., H.Y., and D.S. designed, performed, and analyzed the experiments. D.Y., X.J., H.Y., and D.S. wrote the manuscript. The findings were discussed and the text was reviewed by all authors.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This project was supported by the National Key R&D Program of China (No. 2021YFA160110), National Natural Science Foundation of China (No. 42076212), Innovation Group Projects of Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai) (No. 311021013 and No. 311021011), Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai) (No. SML2021SP311 and No. SML2021SP204), and the Guangzhou Fundamental and Applied Research Project (No. 202201011197).

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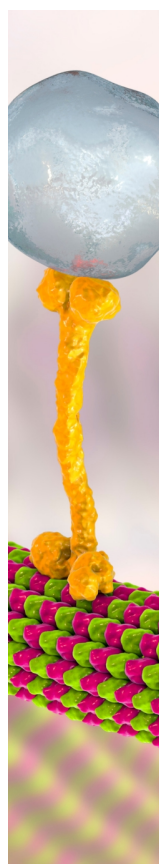
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